

OBERVELLACH, Austria

WMO: 112610

Lat: 46.929N

Lon: 13.223E

Elev: 809

StdP: 91.98

Time Zone: 1.00 (EUC)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-8.3	-15.2	1.1	-5.7	-13.7	1.3	-5.9	6.2	2.5	5.4	2.7	0.6	310	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.1	19.7	27.2	18.8	25.4	18.1	20.3	27.7	19.3	26.0	18.4	24.5	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.9	24.5	16.8	13.2	23.1	16.0	12.6	21.8	62.1	28.0	58.6	26.2	55.5	24.6	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.5	4.0	DB	-12.0	32.5	2.5	2.3	-13.9	34.1	-15.3	35.4	-16.7	36.7	-18.6	38.3	(4)
(5)			WB	-12.5	22.1	2.5	0.7	-14.3	22.6	-15.8	23.0	-17.2	23.4	-19.0	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-0.7	0.6	4.9	9.3	13.5	16.9	18.7	17.9	13.9	9.4	4.3	0.0	(6)	
	DBStd	7.74	3.90	4.06	4.25	3.66	3.37	3.60	3.18	3.27	3.02	3.83	3.46	3.30	(7)	
	HDD10.0	1366	332	264	162	56	7	1	0	0	4	58	172	309	(8)	
	HDD18.3	3481	590	496	416	272	154	70	35	48	134	276	421	568	(9)	
	CDD10.0	1039	0	1	4	34	114	208	271	244	122	41	1	0	(10)	
	CDD18.3	114	0	0	0	0	4	27	47	34	2	0	0	0	(11)	
	CDH23.3	906	0	0	0	3	49	219	369	253	13	0	0	0	(12)	
	CDH26.7	206	0	0	0	0	6	45	85	70	0	0	0	0	(13)	
(14) Wind	WSAvg	1.3	1.1	1.6	1.6	1.6	1.6	1.5	1.4	1.3	1.2	0.9	0.9	1.0	(14)	
(15) Precipitation	PrecAvg	1246	60	53	70	82	108	133	147	144	114	133	136	68	(15)	
	PrecMax	1690	249	231	134	150	219	209	288	215	195	374	400	185	(16)	
	PrecMin	997	9	10	10	23	40	74	48	72	34	7	8	5	(17)	
	PrecStd	178	49	44	34	31	41	38	51	37	42	84	105	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	14.9	19.0	22.6	27.6	29.9	31.1	32.0	25.1	21.3	13.2	8.6	(19)	
		MCWB	5.9	8.6	10.7	14.0	17.9	19.3	20.9	20.8	17.8	15.4	9.2	3.8	(20)	
	2%	DB	8.2	10.9	16.2	20.1	24.8	28.0	29.1	28.5	22.9	18.4	11.3	6.7	(21)	
		MCWB	4.5	5.8	9.5	12.7	16.5	18.9	20.0	19.9	16.8	13.4	8.1	4.1	(22)	
	5%	DB	5.8	8.1	13.9	18.2	22.5	26.0	27.2	26.0	21.0	16.6	10.0	5.4	(23)	
		MCWB	2.5	3.6	8.2	11.4	15.3	18.2	18.8	18.8	15.5	12.4	8.0	2.9	(24)	
	10%	DB	4.0	6.0	11.7	16.2	20.3	23.8	25.3	24.0	19.4	14.9	9.0	4.1	(25)	
		MCWB	1.4	2.2	6.7	10.3	13.8	16.9	18.1	17.6	14.6	11.7	7.4	2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	8.7	11.1	14.4	18.4	20.5	21.7	21.6	18.5	15.5	10.2	6.1	(27)	
		MCDB	10.2	14.6	18.1	21.9	26.2	27.9	29.7	30.4	24.1	20.5	11.5	7.2	(28)	
	2%	WB	4.7	5.8	9.7	13.0	16.9	19.4	20.5	20.3	17.2	14.2	9.3	4.6	(29)	
		MCDB	7.6	10.4	15.7	19.5	24.0	26.5	27.9	27.4	21.8	17.4	10.4	6.0	(30)	
	5%	WB	3.2	4.0	8.3	11.7	15.6	18.5	19.4	19.1	16.1	13.0	8.5	3.4	(31)	
		MCDB	5.0	7.5	13.4	17.3	21.7	25.0	26.1	25.3	19.9	15.7	9.6	4.9	(32)	
	10%	WB	1.9	2.6	7.0	10.6	14.3	17.4	18.4	18.1	15.1	12.0	7.6	2.3	(33)	
		MCDB	3.4	5.4	10.9	15.5	19.6	23.0	24.5	23.2	18.6	14.6	8.7	3.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.8	9.3	9.6	10.3	10.1	10.8	9.8	9.1	7.8	5.6	5.0	(35)	
		MCDBR	7.9	10.2	13.1	13.2	13.8	14.0	13.6	13.3	11.3	10.2	7.4	6.5	(36)	
	5% WB	MCWBR	5.2	6.7	8.1	7.2	7.3	6.7	6.5	6.5	6.1	6.0	5.0	4.6	(37)	
		MCWBR	6.8	9.7	12.6	12.6	13.0	13.1	12.6	12.4	9.9	9.0	5.7	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.295	0.342	0.387	0.402	0.411	0.409	0.402	0.367	0.341	0.306	0.272	(40)	
		taud	2.373	2.300	2.260	2.238	2.265	2.283	2.293	2.330	2.402	2.449	2.478	2.449	(41)	
	Edn at Noon	Edn at Noon	832	872	876	866	864	855	851	842	842	813	781	789	(42)	
		Edh at Noon	70	93	114	129	130	129	126	117	99	81	65	59	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.29	3.35	4.18	4.84	5.41	5.51	4.86	3.60	2.44	1.41	1.17	(44)	
	RadStd	RadStd	0.16	0.30	0.41	0.55	0.59	0.49	0.46	0.52	0.44	0.28	0.25	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		+0.44	N/A	N/A	+0.67	+0.42	+0.33	N/A	-118	+76	+43

Nomenclature: See separate page